

TARGET TOOLCHAINS

GCC 15.2.0 / MSVC 19.51.36257

SPEC

C++17 · Single-Header, 0 Ext. Deps

TARGET ARCHITECTURE

x86_64 · Hardware Invariant TSC

Executive Technical Summary — Independent audit of AnimusCore's lock-free MPMC telemetry ring: zero dynamic allocation post-init, 64-byte cacheline isolation, and hardware-invariant RDTSCP timing verified by compiled, executed harnesses on x86_64. A nanobind zero-copy Python bridge defect was found and patched live this cycle; all figures below are measured, not asserted.

CRITICAL MEMORY TOPOLOGY & CACHELINE ISOLATION

```
┌ MpmcBoundedQueue<Event> — pow2 capacity, heap-allocated once at construction ─────────┐  
│ Cell[i] = { atomic<u64> sequence (CAS guard, per-slot) | Event data (64B payload) }      │  
│                                                                                          │  
│ alignas(64) write_head — cacheline A (producer cursor, CAS-owned)                      │  
│ alignas(64) read_tail  — cacheline B (consumer cursor, CAS-owned) ← separated: no false-share │  
│                                                                                          │  
│ struct alignas(64) Event { u64 seq, dispatch_tsc; u32 producer_id,_pad; u64 value; u64 _pad[4]; }  
│ sizeof(Event)=64 alignof(Event)=64 is_trivially_copyable<Event>=true (static_assert) │  
└──────────────────────────────────────────────────────────────────────────────────────────┘
```

CORE INVARIANT TABLE

INVARIANT	TARGET	MEASURED	STATUS
Dynamic heap, post-init	0 B	0 B – 7 configs (1P/1C+8P/8C, Release+ASan) + 51.6B-event soak, all 0	PASS
Cacheline contention	Isolated	1.29–1.31× throughput vs. unpadded (2 pinned threads, 20M iters/thread, 3 consistent runs)	PASS
Cycle counting	0 kernel switches	RDTSCP, invariant-TSC confirmed (CPUID 0x80000007:EDX[8]), 10.16 ns/read, 2.4192 cyc/ns calib.	PASS
Lock-free enqueue/dequeue	No mutex/syscall	static_assert atomic::is_always_lock_free; CAS-retry only (source-verified)	PASS

SUBSYSTEM INTEGRATION MATRIX

SUBSYSTEM	DETAIL
Header footprint	ring_buffer.hpp + sandbox_event.hpp + tsc_clock.hpp — ~250 lines total, 0 external dependencies, C++17 only.
Python interop bridge	nanobind TelemetryStream::drain() bound with nb::rv_policy::reference_internal — attaches the owning instance as the returned ndarray's owner (nanobind's built-in equivalent of a capsule-based owner handle), aliasing the C++ ring's contiguous scratch buffer directly into NumPy with no per-call heap copy . Measured 25.4–33.2 ns/event . Patched and re-verified this audit cycle — see Page 2 §3.

SOAK STRESS TEST MATRIX — 51.6B EVENTS / 2H 38M (HISTORICAL, 2026-09-14)

TARGET METRIC	TARGET	MEASURED	STATUS
Events produced = consumed	Exact match	51,623,591,903 = 51,623,591,903	PASS
Drop deficit	0	0 — exact accounting, whole run	PASS
Memory (heap) deficit	0 B, hot path	0 B — tripwire armed post-init, entire 9,529.1s	PASS
ASan / UBSan status	Clean	Clean per 2026-09-14 log; MSVC ASan re-run this cycle (4P/4C, 8P/8C) also clean, 0 faults	PASS
Duration / config	—	9,529.1 s (2h 38m 49s) continuous, 8P/8C, 1,048,576-slot ring	—

Throughput — Concurrency Scaling (fresh, Release, this cycle)

PAIRS	THROUGHPUT	P50
1P / 1C	27.3M/s	82 ns
2P / 2C	17.0M/s	~230 ns
4P / 4C	9.9M/s	~690 ns
8P / 8C	7.9M/s	~2.0 μs

Independently corroborated: 7.26–7.29M pushes/sec (±1%) over a separate 40s sustained 8P/8C run — see WHITEPAPER.md.

Latency Spectrum — Read Separately, Not Blended

SINGLE-TICK PINNED LATENCY (1P/1C)

P50: 82 ns

Sub-100ns hot path confirmed — dedicated cores, Release build, low contention

SATURATED MULTI-THREAD QUEUE BACKLOG (8P/8C, 2.65H SUSTAINED)

P50: ~134 ms

Oversubscription backlog on 24T host over 2.65h — not an algorithmic floor. Never cite interchangeably with the pinned figure above.

LIVE BUG DIAGNOSTIC & PATCH VERIFICATION — ZERO-COPY BRIDGE

ITEM	DETAIL
Defect	nanobind drain() returned a fresh copy every call, not a view — .ctypes.data alternated between 2 addresses across 49 calls.
Root cause	Default rv_policy::automatic copies whenever the ndarray has no owner and no self (nanobind nb_ndarray.cpp, source-verified).
Patch	Bound drain() with nb::rv_policy::reference_internal — attaches TelemetryStream as owner; 1-line binding change, no C++ function-body change.
Verification	CONFIRMED — pointer trace: 1 stable address / 49 calls (post-fix) vs. 2 alternating (pre-fix); lifetime-safety test (owner GC'd mid-view) passed; perf unchanged, 25.4 ns/event median.

INSTITUTIONAL DEPLOYMENT & EVALUATION TERMS

PROOF-OF-CONCEPT

48-Hour Sandbox

Self-contained animus_sandbox/, 2-command build, 0 external deps, reproduces every figure above on your own hardware.

LICENSING MODEL

Desk → Enterprise

Desk License \$30–60K/yr (single node) · Institutional Enterprise \$240K/yr (multi-node, one desk).

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